Plan of Day

- Reminders
  - Next week is Spring Break– no class
  - Required Assignments Reminder
  - What's my grade tool

  - Open questions: Investments
  - Today: Discussions on Debt

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CWRU
Essentials of Personal Finance
Week 8

Managing and Paying Debt
Home and School Debt

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How Debt Works

Borrowing Money is not free --- Interest!

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Repaying Debt

Amount Borrowed	10,000							
Borrowing Rate	5%							
N Periods	120	(10 years = 120 months)						
Payment	$106.07							
	Beginning Balance	Interest	Adjusted Balance	Required Payment	Extra Payment	Total Payment	Payment applied to Principal	Ending Balance
Month 1	10,000	41.66667	10,042	$106.07	0	$106.07	$64.40	$9,935.60
Month 2	$9,935.60	41.39834	9,977	$106.07	0	$106.07	$64.67	$9,870.93
Month 3	$9,870.93	41.12889	9,912	$106.07	0	$106.07	$64.94	$9,806.00
Month 4	$9,806.00	40.85832	9,847	$106.07	0	$106.07	$65.21	$9,740.79
Month 5	$9,740.79	40.58663	9,781	$106.07	0	$106.07	$65.48	$9,675.31
Month 6	$9,675.31	40.3138	9,716	$106.07	0	$106.07	$65.75	$9,609.56
Month 7	$9,609.56	40.03983	9,650	$106.07	0	$106.07	$66.03	$9,543.53
Month 8	$9,543.53	39.76472	9,583	$106.07	0	$106.07	$66.30	$9,477.23
Month 9	$9,477.23	39.48847	9,517	$106.07	0	$106.07	$66.58	$9,410.66
Month 10	$9,410.66	39.21107	9,450	$106.07	0	$106.07	$66.85	$9,343.80
Month 11	$9,343.80	38.93251	9,383	$106.07	0	$106.07	$67.13	$9,276.67
Month 12	$9,276.67	38.65279	9,315	$106.07	0	$106.07	$67.41	$9,209.26
Months Remaining	108					Total Principal		$ 10,000
In Years	9					Total Interest		$ 2,728
						Total Payments		$ 12,728

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Amount Borrowed	10,000							
Borrowing Rate	5%							
N Periods	120	(10 years = 120 months)						
Payment	$106.07							
Extra Payment	50							
	Beginning Balance	Interest	Adjusted Balance	Required Payment	Extra Payment	Total Payment	Payment applied to Principal	Ending Balance
Month 1	10,000	41.66667	10,042	$106.07	50	$156.07	$114.40	$9,885.60
Month 2	$9,885.60	41.19	9,927	$106.07	50	$156.07	$114.88	$9,770.73
Month 3	$9,770.73	40.71136	9,811	$106.07	50	$156.07	$115.35	$9,655.37
Month 4	$9,655.37	40.23071	9,696	$106.07	50	$156.07	$115.83	$9,539.54
Month 5	$9,539.54	39.74807	9,579	$106.07	50	$156.07	$116.32	$9,423.22
Month 6	$9,423.22	39.26341	9,462	$106.07	50	$156.07	$116.80	$9,306.42
Month 7	$9,306.42	38.77674	9,345	$106.07	50	$156.07	$117.29	$9,189.13
Month 8	$9,189.13	38.28803	9,227	$106.07	50	$156.07	$117.78	$9,071.35
Month 9	$9,071.35	37.7973	9,109	$106.07	50	$156.07	$118.27	$8,953.08
Month 10	$8,953.08	37.30451	8,990	$106.07	50	$156.07	$118.76	$8,834.32
Month 11	$8,834.32	36.80967	8,871	$106.07	50	$156.07	$119.26	$8,715.07
Month 12	$8,715.07	36.31277	8,751	$106.07	50	$156.07	$119.75	$8,595.31
If continue the extra payments						Total Principal		$ 10,000
Months Remaining	63					Total Interest		$ 1,657
In Years	5					Total Payments		$ 11,657
						Interest Saved:		$ 1,070
						Time (months) saved		45
						Time (years) saved		4

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Amount Borrowed	100,000							
Borrowing Rate	5%							
N Periods	120	(10 years = 120 months)						
Payment	$1,060.66							
	Beginning Balance	Interest	Adjusted Balance	Required Payment	Extra Payment	Total Payment	Payment applied to Principal	Ending Balance
Month 1	100,000	416.6667	100,417	$1,060.66	0	$1,060.66	$643.99	$99,356.01
Month 2	$99,356.01	413.9834	99,770	$1,060.66	0	$1,060.66	$646.67	$98,709.34
Month 3	$98,709.34	411.2889	99,121	$1,060.66	0	$1,060.66	$649.37	$98,059.97
Month 4	$98,059.97	408.5832	98,469	$1,060.66	0	$1,060.66	$652.07	$97,407.90
Month 5	$97,407.90	405.8663	97,814	$1,060.66	0	$1,060.66	$654.79	$96,753.11
Month 6	$96,753.11	403.138	97,156	$1,060.66	0	$1,060.66	$657.52	$96,095.60
Month 7	$96,095.60	400.3983	96,496	$1,060.66	0	$1,060.66	$660.26	$95,435.34
Month 8	$95,435.34	397.6472	95,833	$1,060.66	0	$1,060.66	$663.01	$94,772.33
Month 9	$94,772.33	394.8847	95,167	$1,060.66	0	$1,060.66	$665.77	$94,106.56
Month 10	$94,106.56	392.1107	94,499	$1,060.66	0	$1,060.66	$668.54	$93,438.02
Month 11	$93,438.02	389.3251	93,827	$1,060.66	0	$1,060.66	$671.33	$92,766.69
Month 12	$92,766.69	386.5279	93,153	$1,060.66	0	$1,060.66	$674.13	$92,092.56
Months Remaining	108					Total Principal		$ 100,000
In Years	9					Total Interest		$ 27,279
						Total Payments		$ 127,279

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Amount Borrowed	100,000							
Borrowing Rate	5%							
N Periods	120	(10 years = 120 months)						
Payment	$1,060.66							
Extra Payment	500							
	Beginning Balance	**Interest**	**Adjusted Balance**	**Required Payment**	**Extra Payment**	**Total Payment**	**Payment applied to Principal**	**Ending Balance**
Month 1	100,000	416.6667	100,417	$1,060.66	500	$1,560.66	$1,143.99	$98,856.01
Month 2	$98,856.01	411.9	99,268	$1,060.66	500	$1,560.66	$1,148.76	$97,707.26
Month 3	$97,707.26	407.1136	98,114	$1,060.66	500	$1,560.66	$1,153.54	$96,553.71
Month 4	$96,553.71	402.3071	96,956	$1,060.66	500	$1,560.66	$1,158.35	$95,395.37
Month 5	$95,395.37	397.4807	95,793	$1,060.66	500	$1,560.66	$1,163.17	$94,232.19
Month 6	$94,232.19	392.6341	94,625	$1,060.66	500	$1,560.66	$1,168.02	$93,064.17
Month 7	$93,064.17	387.7674	93,452	$1,060.66	500	$1,560.66	$1,172.89	$91,891.28
Month 8	$91,891.28	382.8803	92,274	$1,060.66	500	$1,560.66	$1,177.77	$90,713.51
Month 9	$90,713.51	377.973	91,091	$1,060.66	500	$1,560.66	$1,182.68	$89,530.83
Month 10	$89,530.83	373.0451	89,904	$1,060.66	500	$1,560.66	$1,187.61	$88,343.22
Month 11	$88,343.22	368.0967	88,711	$1,060.66	500	$1,560.66	$1,192.56	$87,150.66
Month 12	$87,150.66	363.1277	87,514	$1,060.66	500	$1,560.66	$1,197.53	$85,953.13
If continue the extra payments....						Total Principal		$ 99,977
Months Remaining	63					Total Interest		$ 16,575
In Years	5					Total Payments		$ 116,552
						Interest Saved:		$ 10,704
						Time (months) saved		45
						Time (years) saved		4

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Managing Debt: 5 Factors

1. Your Current Debt Load
2. Your Current Cash-flow and Savings
3. Your Current Tax Status
4. Your Age
5. Your Psychological Relationship with Debt

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Your Current Debt Load:

How much do you owe???

1. Create a list of all your outstanding debt
   - Auto, Home, Student Loans, Credit Cards, Any debt you guaranteed/co-signed.
2. List the Interest Rate (the borrowing cost) of each form of Debt
   - Look for the Annual Percentage Rate ("APR") listed on each document
3. Determine tax deductibility of each type of interest. Determine after-tax cost of borrowing if deductible.
   - Mortgage Debt – You must itemize to get deduction (consider new tax law)
   - Student Loans – You can take standard deduction or itemize

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List your debt

- Example: (assumes taxpayer is in 15% tax bracket)

Description / Lender	APR / Cost %	Deductible?	After-tax Cost %
1. Student Loan	5.5%	Yes, if below AGI limits	5.5 – (5.5*.15) = 4.65%
2. Credit Card	12%	No	12%
3. Home Mortgage	4.5%	Yes, if itemize	4.5% (standard deduction) or 4.5 – (4.5*.15) = 3.83%

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List your debt

- Determine if each debt can be cancelled in Bankruptcy.
  - While much debt can be cancelled in bankruptcy, some debt cannot be eliminated in bankruptcy;
    - Student Loan Debt (there are some rare circumstances where it can)
    - Past Due Taxes
- DO NOT RELY ON BANKRUPTCY TO ELIMINATE DEBT - It's a last resort, an extreme solution, but knowing what can be eliminate in bankruptcy will help prioritize order of debt pay down.
- Filing for Bankruptcy will have severe implications for your credit score.

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Which Debt to Pay First

- Pay debt with highest after-tax cost first
- If you have a "tie" between 2 loans:
  - Pay what cannot be written-off in bankruptcy first, then what can be written off in bankruptcy.

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Receive unexpected cash?

- Add to my savings?
- Pay down debt?
- Spend it?

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Are you able to service current debt?

- Paying your outstanding debt in a timely manner?
  - Making more than minimum monthly payments?
  - Are you current on ALL of your outstanding debt?
    If yes, then consider saving ALL or Part of the new cash.
- Having trouble paying your current debt?
  - Use ALL or a MAJORITY of the new cash to PAY DOWN DEBT
  - Pay debt with highest interest rate first
  - Do NOT borrow any new money until you retire all your debt.

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Your Age Impacts Debt Pay down

- If you are below 40 – INVEST!!!!!
  - You still have many years left before you retire.
  - Compound interest is your "best friend"
  - You can afford temporary dips in the financial markets (i.e. you have time to allow the market to recover)
  - You should be able to earn a higher rate of return on investments than your cost of debt
- If you are between 40 and 60 – It Depends!
  - The importance of hitting your savings goal are becoming much more important
  - If you are hitting your savings goals, you have the option to start paying off debt early
  - If you are not hitting your goals, you should NOT pay-off debt early, you need to save for retirement.

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Your Age Impacts Debt Pay down

- If you are over 60 – PAY OFF DEBT!
  - You need to pay-off your mortgage BEFORE you retire.
  - You are reducing the risk in your investment portfolio, which means your annual average rate of return will be lower.
    - Consequently, you may not be able to earn a higher rate of return on investments than your cost of debt.
  - At this state, the standard deduction will most likely offer more benefit than itemized tax deductions.

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Your Psychological Relationship with Debt

- Risk Aversion Level
  - Some people sleep extremely well owing debt
    - Accept paying the cost to carry debt
    - Confident of their future ability to pay off the debt and maintain an adequate savings program.
  - Some people can't sleep well owing debt
    - Do not like paying interest on debt
    - Worried about their future ability to retirement debt and maintain an adequate savings program.

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Mortgages

Home Loan Rules of Thumb

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But first….Home Buying Tips

- A home is both consumption and investment (in my opinion).
  - Consider Resale Value
  - For ROI, plan to own for at least 5 years
  - Understand Potential Tax Exclusion
- Know what you can afford
  - Check your credit score: Higher score = lower borrowing rate
  - Downpayment (I recommend 20%)
  - Understand Other Costs
    - Property Tax
    - Insurance
    - Maintenance
    - HOA, Condo fees, etc.
- Get Pre-approved (this is the bank's opinion of your MAX)
  - Don't look at homes above what you can afford
  - Just because you can, doesn't mean you should

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Mortgage Debt

Example
Salary $60,000
Monthly $5,000
28% $1,400
35% $1,750

- What the bank thinks: "Ideal Mortgage"
  - Monthly housing costs (Mortgage payment, Insurance, Prop tax) should not exceed 28% of pre-tax monthly income
  - Monthly installment debt costs (housing + credit card + auto + student loan) should NOT exceed 35% of pre-tax monthly income.
  - If you borrow for 30 years:
    - 1st mortgage payment - 92% interest, 7% principal
    - Takes approximately 24 years before 50% of mortgage payment is applied against principal.

Just because you can, doesn't mean you should!
Avoid looking at houses above your budget!

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Recommendations (Mortgage Debt)

- You need to conserve cash flow for savings
- Keep your total monthly housing costs at no more than 20% of pre-tax monthly income
- Keep installment debt monthly costs at no more than 30% of pre-tax monthly income
- Avoid borrowing more than 2.5x your annual income
  - Median home value / Median household income in 2023 =6.06x.
  - Be Unique! Be Smart!

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Recommendations (Mortgage Debt)

- When you buy a home with a mortgage, the bank owns the home.
- Borrow for 30 years, but never take 30 years to pay it off – aim for paying off in 20 years.
  - Maximize your financial flexibility
  - Reduce the cost of paying off the mortgage

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Recommendations (Mortgage Debt)

- Example:
- The monthly payment on a $140,000, 30 year mortgage at a fixed rate of 5%:
  - PV = -140,000, n= 360, I = 5/12 or .41%, PMT = $751.55
  - Total cost of Loan = $751.55 x 360 = $270,558
- Impact of extra $150 monthly payment against outstanding principal
  - PV= -140,000, I = 5/12 or .41%, PMT = 751.55 +150 =901.55
  - Solve for n = 250.5 or 20.8 years!
  - Total cost of Loan = $902.55 x 250.5 = $226,008
    - Difference: $44,550

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Recommendations (Mortgage Debt)

- Temporarily lower your mortgage payment
  - If you have a financial emergency, you can only make the "required payment" of $751.55 until your situation improves
- Reduces your cost to purchase a house by 16.9%
  - You can save the difference between total paid back over 30 years and total paid back over 20.5 years of $44,550.
- Get the biggest raise you will EVER receive
  - Once you pay-off the mortgage (after 20.3 years), your monthly after-tax income increases by $901.55

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STUDENT LOANS

Dealing with Student Loan Debt

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Student Loans "The new reality"

- Student Loans are FOREVER!!!
  - Cannot be extinguished in bankruptcy
  - Most "guaranteed student loan debt" cannot be extinguished by death.
    - Consider term life insurance to manage this risk
  - Federal student loans – most are forgiven if borrower dies

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Subsidized versus Unsubsidized Loan

- Subsidized – No interest accrued while you are in school.

Amount Borrowed		10000		
Borrowing Rate		6%		
Subsidized		Yes		
	Beginning Loa	Interest	Payment	Ending Loan
Year 1	$ 10,000	$ -	$ -	$ 10,000
Year 2	$ 10,000	$ -	$ -	$ 10,000
Year 3	$ 10,000	$ -	$ -	$ 10,000
Year 4	$ 10,000	$ -	$ -	$ 10,000

- Unsubsidized Loan – Interest accrues while you are in school. Even though no payments are required, interest accrues and increases the amount of your debt.

Amount Borrowed		10000		
Borrowing Rate		6%		
Subsidized		No		
	Beginning Loa	Interest	Payment	Ending Loan
Year 1	$ 10,000	$ 600	$ -	$ 10,600
Year 2	$ 10,600	$ 636	$ -	$ 11,236
Year 3	$ 11,236	$ 674	$ -	$ 11,910
Year 4	$ 11,910	$ 715	$ -	$ 12,625

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Recommendations (Student Loans)

- Studies show that students with some responsibility in paying for higher education (including student loans) take it more seriously
  - It **may** make sense to help your child AFTER college to pay-down student loans rather than avoid taking out the student loan during education (consider timing of when you discuss such plan with child)
- Shoot to pay off student loan debt in 10 years, but if qualify for a forgiveness program, then a longer payback period may be more appropriate
  - Avoid borrowing more than 1 year of salary expected post-graduation

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Recommendations (Student Loans)

- Learn about student loan forgiveness programs
- Learn about employer provided student loan assistance programs (Fairly New Employee benefit plan)
- Consider consolidating / refinancing student loans for lower interest rates (but keep payment the same to pay off loan quicker)
  - Look at Direct Consolidation Loans first to maintain ability to participant in an income-based repayment plan

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Student Loan Repayment

- Standard Payback Plan – 10 year fixed
- Income Based Repayment Plans
  - Federal loans allow for income-based repayment plans
  - Monthly loan payment is based on a percentage of "Disposable Income"
    - Lower monthly loan payments
    - Loan Forgiveness (Remaining Balance is Forgiven!) after 20-25 years of monthly payments
  - May be more costly way to borrow over the long-term
    - While your monthly payments are lower, your outstanding loan balances accumulates much more interest than under a 10 year repayment term

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Student Loan Debt Forgiveness

- Debt forgiven will be TAXED as ORDINARY INCOME in year of forgiveness.
  - Borrower will owe income tax on that income
    - Tax liability should not be the deciding factor.
    - Run the numbers
  - Exception: Borrower qualifies for a Public Service Loan Forgiveness Program.

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Student Loans
Income Driven Repayment Plans

Repayment Plan	Monthly Payment Amount	Repayment Period
Income-Based (IBR)	15% of discretionary income. (10% for new borrowers) The payment will never be more than the amount you would pay under the 10-year Standard Repayment Plan.	25 years (20 years for new borrowers).
Pay As You Earn (PAYE)	10% of discretionary income The payment will never be more than the amount you would pay under the 10-year Standard Repayment Plan.	20 years
Revised Pay As You Earn (REPAYE)	10% of discretionary income	20 years if all loans being repaid on the plan were received for undergraduate study 25 years if any loans being repaid under the plan were received for graduate or professional study
Income-Contingent (ICR)	The lesser of: 20% of discretionary income, or monthly payment on a 12-year fixed plan, adjusted based on your income.	25 years

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Resources

- http://cfs.cuyahogacounty.us/en-US/Financial-Literacy-Money-Management.aspx
  - Links to organizations with free services
- Consumer Credit Counseling Services
  - Free Credit Counseling Sessions
  - Free Debt Specific Seminars
  - Debt Management Plans have fees
- Credit counseling services differ from debt settlement services
  - Beware of debt settlement companies that charge up-front fees in return for promising to settle your debts

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QUESTIONS

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Use the below space to make a list of your current debts

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